

Report writing format:

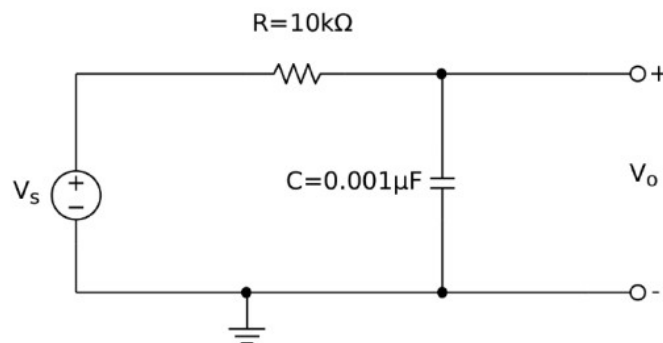
- 1. Objectives**
 - 2. Introduction**
 - 3. Results& Analysis (answer the following problems in this section) &**
 - 4. Conclusion & Discussion**
- Submission Format: pdf file.**

This lab has 2 parts:

1. Low Pass Filter Response.
2. High Pass Filter Response

➤ **Experiment #1: Low Pass Filter Response**

Construct the following circuit, where V_s is an AC signal (sinusoidal voltage source) having an amplitude of 10 V or 20 V_{p-p}.



1. Measure and record the actual values of the components using an ohmmeter and capacitance meter.

R (actual) = _____ C (actual) = _____

2. Using the oscilloscope, measure both the input and the output voltages (magnitude and phase difference) as a function of frequency for the different frequencies shown in the Table below.

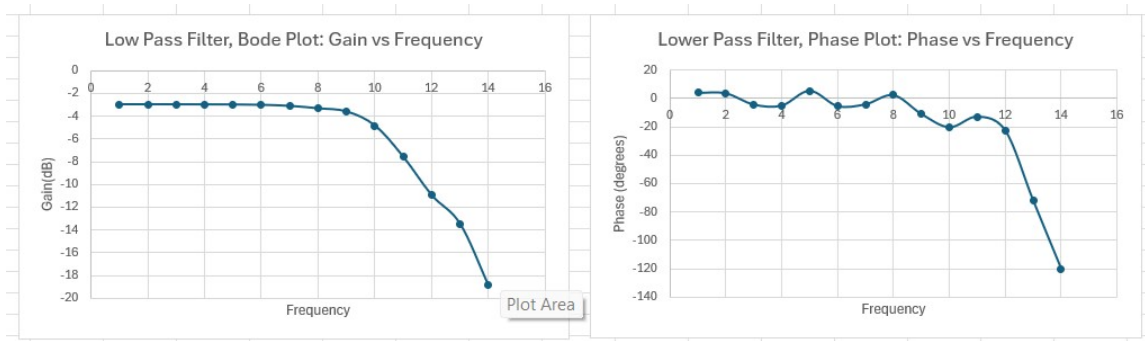
Fill in the table below.

R=10KΩ C=0.001μF

R(actual) = 9.8824KΩ C(actual)=209.7pF

CH1 on resistor and CH2 on capacitor for Low Pass filter

Suggested Frequency	Actual frequency	Vs	Vo	Phase	Vo/Vs	(Vo/Vs) dB			
100 Hz	100Hz	10Vpp	7.108	3.96	0.7108	-2.96505			
200 Hz	200Hz	10Vpp	7.107	3.61	0.7107	-2.96627			
400 Hz	400Hz	10Vpp	7.104	-4.32	0.7104	-2.96994		Low Pass Filter	
700 Hz	700Hz	10Vpp	7.099	-5.03	0.7099	-2.97606			
1 KHz	1 KHz	10Vpp	7.093	5.4	0.7093	-2.9834			
2 KHz	2 KHz	10Vpp	7.067	-5.4	0.7067	-3.0153			
4 KHz	4 KHz	10Vpp	6.992	-4.11	0.6992	-3.10797			
7 KHz	7 KHz	10Vpp	6.822	2.29	0.6822	-3.32177			
10 KHz	10 KHz	10Vpp	6.607	-11	0.6607	-3.59991			
20 KHz	20 KHz	10Vpp	5.728	-20.3	0.5728	-4.83994			
40 KHz	40 KHz	10Vpp	4.175	-12.8	0.4175	-7.58687			
70 KHz	70 KHz	10Vpp	2.836	-22.6	0.2836	-10.9459			
100 KHz	100 KHz	10Vpp	2.123	-72	0.2123	-13.461			
200 KHz	200 KHz	10Vpp	1.139	-120	0.1139	-18.8695			





Include the above Table, draw two graphs, and add in your report.

3.

- a) From the Bode plot, resulting from the measured data estimate the 3 dB frequency (f_{3dB}) and report its value in your report.

$$f_{3dB} \text{ (measured)} = \underline{2 \text{ KHz}}$$

- b) From the measured values of the actual R and C components calculate the f_{3dB} and report its value in your report.

$$f_{3dB} \text{ (calculated)} = 76.799 \text{ KHz}$$

Low Pass Filter

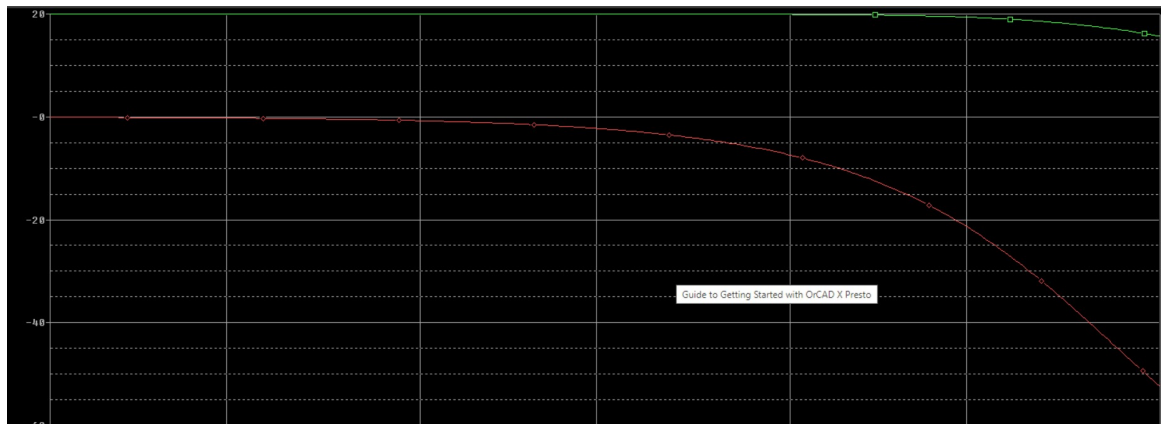
$$f_{3dB} = \frac{1}{2\pi RC}$$

$$f_{3dB} = \frac{1}{2\pi (9.8824 \text{ k}\Omega) \cdot (209.7 \text{ pF})}$$

$$f_{3dB} = 76.799 \text{ kHz}$$

- c) Simulate the frequency response of this circuit with Pspice. Use an AC voltage source and perform an AC sweep of the frequency in the range 100 Hz to 1 MHz. Include the PSpice circuit and the Bode plot generated by the PSpice simulation in your lab report. Determine the f_{3dB} as shown in the lecture.

$$f_{3dB} (\text{PSpice}) = 75.71661 \text{ KHz}$$



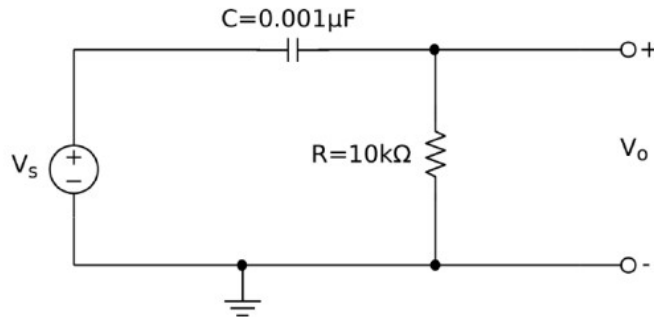
	Evaluate	Measurement	Value
☒	☒	Cutoff_Lowpass_3dB(V(C2:2))	75.71661k
☐	☐		

In your report, compare the three f_{3dB} values. Explain any differences.

There are some big differences in my measured vs calculated/pspice values. My measured value was 2KHz which is way smaller than calculated value=76.799KHz/psice value=75.71661KHz. My calculated and pspice values are within range, but my measured values are way off. I think this could be caused by equipment error because my 0.001 microfarad capacitor actual reading was 209.7 picofarad's. This is way off the capacitor's actual value, in part 2 of the experiment my 0.001 microfarad capacitor's reading was 934.5 picofarad's. My data in part 2 has a closer correlation with each other.

➤ Experiment #2: High Pass Filter Response

Construct the following circuit, where V_s is an AC signal (sinusoidal voltage source) having an amplitude of 10 V or 20 V_{p-p}.



1. Measure and record the actual values of the components using an ohmmeter and capacitance meter, if the components are different than those from experiment #1.

R (actual) = _____ C (actual) = _____

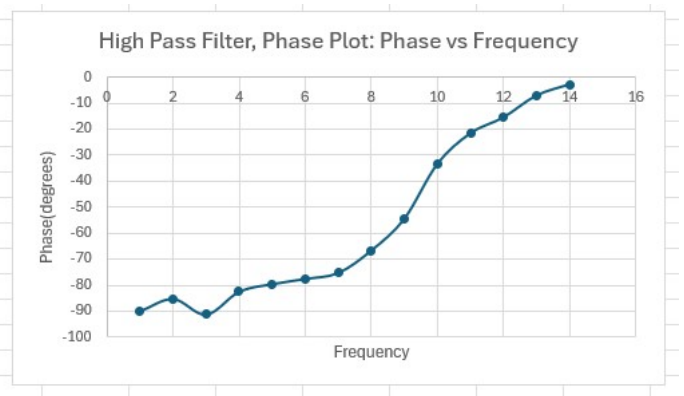
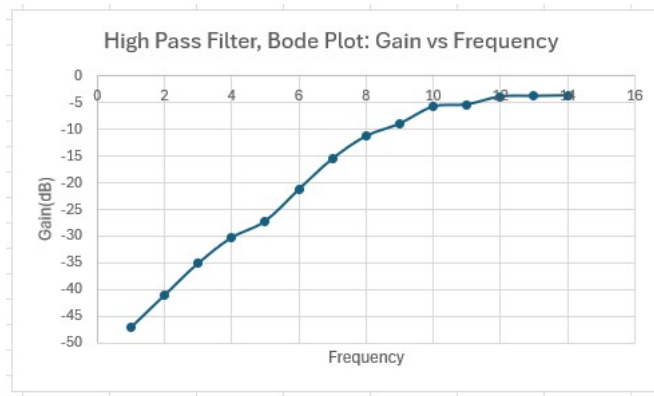
2. Using the oscilloscope, measure both the input and the output voltages (magnitude and phase difference) as a function of frequency for the different frequencies shown in the Table below. Fill in the table below.

R=10KΩ C=0.001μF

R(actual) = 9.8508KΩ C(actual)=905.9pF

CH1 on capacitor and CH2 on resistor for High Pass filter.

Suggested Frequency	Actual Frequency	Vs	Vo	Phase	Vo/Vs	(Vo/Vs) dB		
100 Hz	100 Hz	10Vpp	.044066V	-90	0.004407	-47.1179		
200 Hz	200 Hz	10Vpp	.088092V	-85.3	0.008809	-41.1013		
400 Hz	400 Hz	10Vpp	0.17557V	-91.3	0.017557	-35.111		
700 Hz	700 Hz	10Vpp	0.30632V	-82.5	0.030632	-30.2765		
1 KHz	1 KHz	10Vpp	0.43648V	-79.6	0.043648	-27.1936		
2 KHz	2 KHz	10Vpp	0.86202V	-77.6	0.086202	-21.2897		
4 KHz	4 KHz	10Vpp	1.6702V	-75.3	0.16702	-15.5446		
7 KHz	7 KHz	10Vpp	2.7242V	-66.7	0.27242	-11.2952		
10 KHz	10 KHz	10Vpp	3.5653V	-54.5	0.35653	-8.95808		
20 KHz	20 KHz	10Vpp	5.1541V	-33.5	0.51541	-5.75694		
40 KHz	40 KHz	10Vpp	6.0978V	-21.7	0.60978	-5.34971		
70 KHz	70 KHz	10Vpp	6.4051V	-15.6	0.64051	-3.86948		
100 KHz	100 KHz	10Vpp	6.4904V	-7.21	0.64904	-3.75457		
200 KHz	200 KHz	10Vpp	6.5680V	-2.91	0.6568	-3.65134		



Include the above Table, draw two graphs and add in your report

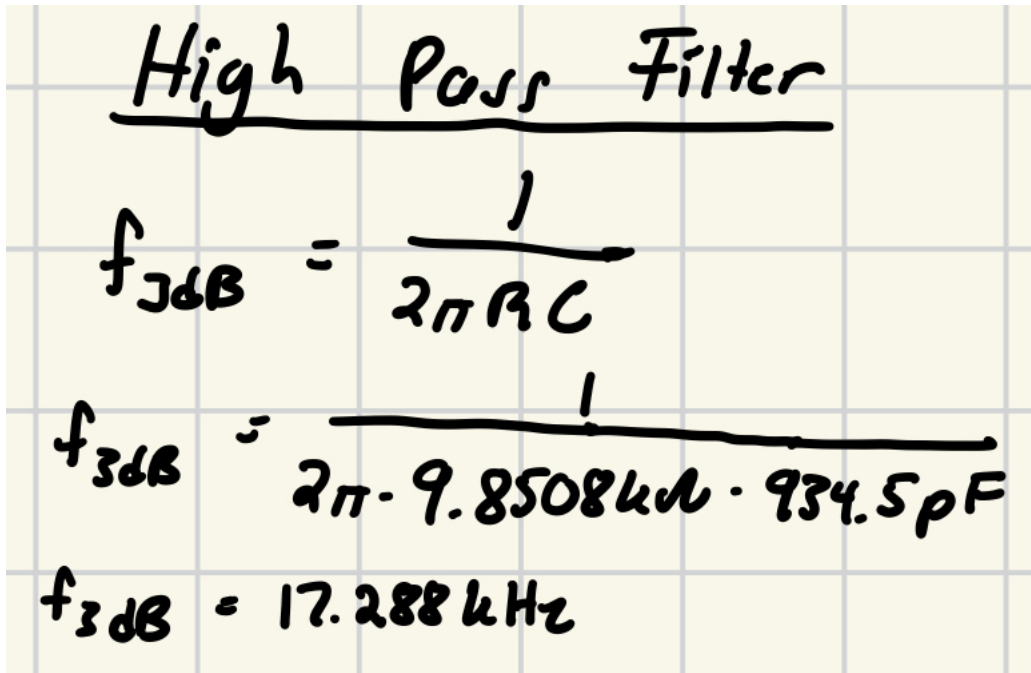
4.

- d) From the Bode plot, resulting from the measured data estimate the 3 dB frequency (f_{3dB}) and report its value in your report.

$$f_{3dB} \text{ (measured)} = \underline{20\text{KHz}}$$

- e) From the measured values of the actual R and C components calculate the f_{3dB} and report its value in your report.

$$f_{3dB} \text{ (calculated)} = \underline{17.288\text{KHz}}$$

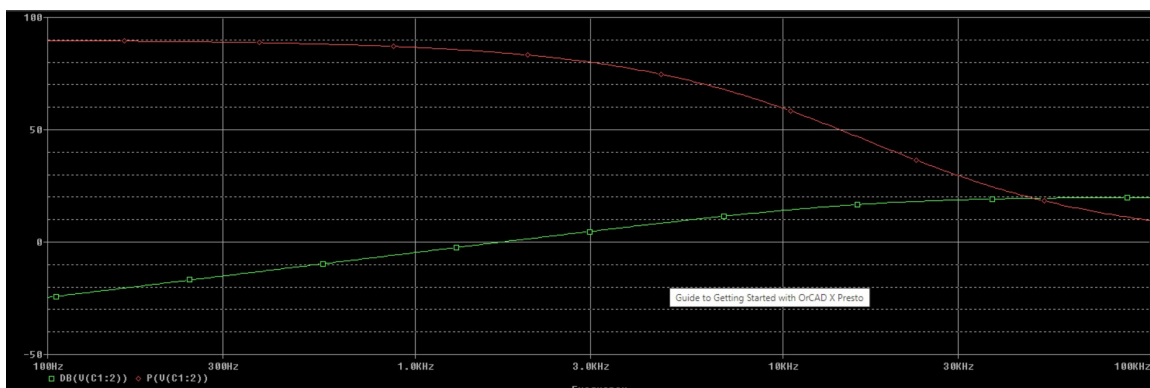


Handwritten calculation for the cutoff frequency of a High Pass Filter:

$$f_{3dB} = \frac{1}{2\pi RC}$$
$$f_{3dB} = \frac{1}{2\pi \cdot 9.8508\text{k}\Omega \cdot 934.5\text{pF}}$$
$$f_{3dB} = 17.288\text{kHz}$$

- f) Simulate the frequency response of this circuit with PSpice. Use an AC voltage source and perform an AC sweep of the frequency in the range 100 Hz to 1 MHz. Include the PSpice circuit and the Bode plot generated by the PSpice simulation in your lab report. Determine the f_{3dB} as shown in the lecture.

$$f_{3dB} \text{ (PSpice)} = \underline{16.59582\text{KHz}}$$



	Evaluate	Measurement	Value
☐	☐	Cutoff_Lowpass_3dB(V(C2:2))	
☒	☒	Cutoff_Highpass_3dB(V(R1:2))	16.59582k
☐			

In your report, compare the three f_{3dB} values. Explain any differences.

My measured, calculated, and pspice values are all within range of each other. Measured Value = 20KHz, Calculated Value=17.288KHz, and Pspice Value=16.59582KHz. All of these values are way closer than part a of the experiment. The reasoning for the higher measured value could be from measuring equipment error.